Bridgeland Stability Conditions

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Contents

1. Stability conditions on an abelian category 1
1.1. Conventions 1
1.2. Stability functions 2
1.3. Harder–Narasimhan filtrations and stability conditions 2
1.4. Basic properties 3
1.5. Uniqueness of Harder–Narasimhan filtrations 5
1.6. Existence of Harder–Narasimhan filtrations 6
1.7. Examples 12
2. t-structures and their hearts 14
2.1. Conventions on triangulated categories 14
2.2. Definition 15
2.3. Truncation functors 15
2.4. The heart 17
References 20

1. Stability conditions on an abelian category

1.1. **Conventions.** This section follows [MS17, §4]; the notion of a stability function on an abelian category goes back to Rudakov [Rud97]. Throughout, $\mathcal{A}$ is an essentially small abelian category and $K_0(\mathcal{A})$ is its Grothendieck group, i.e. the abelian group generated by the classes $[E]$ of objects $E \in \mathcal{A}$ subject to the relations $[E] = [E'] + [E'']$ for every short exact sequence $0 \to E' \to E \to E'' \to 0$. For an object E and a group homomorphism $Z\colon K_0(\mathcal{A}) \to \mathbb{C}$ we write $Z(E) := Z([E])$.

Let

$$\mathbb{H} := \left\{ re^{i\pi\varphi} \middle| \, r > 0,\ \varphi \in (0,1] \right\} = \{ z \in \mathbb{C} | \operatorname{Im} z > 0 \} \cup \mathbb{R}_{<0}$$

be the upper half-plane together with the negative real ray. For $z \in \mathbb{H}$ the number $\varphi \in (0,1]$ with $z \in \mathbb{R}_{>0} e^{i\pi\varphi}$ is uniquely determined; we call it the *phase* of z. Note that $\mathbb{H}$ is closed under addition and under multiplication by positive real numbers.

The notation $F \subset E$ and $E \twoheadrightarrow G$ always refers to a subobject and a quotient object in the abelian category $\mathcal{A}$. A subobject $F \subset E$ is *proper* if $F \neq E$ and *nontrivial* if $F \neq 0$.

1.2. **Stability functions.** A group homomorphism $Z\colon K_0(\mathcal{A}) \to \mathbb{C}$ is called a *stability function* on $\mathcal{A}$ ([MS17, Def. 4.1], [Bri07, Def. 2.1]) if

(Z) $Z(E) \in \mathbb{H}$ for every nonzero object $E \in \mathcal{A}$.

Equivalently: $\operatorname{Im} Z(E) \geqslant 0$, and $\operatorname{Im} Z(E) = 0$ implies $\operatorname{Re} Z(E) < 0$.

Let Z be a stability function. For a nonzero object $E \in \mathcal{A}$ the *phase* of E is the number $\varphi(E) = \varphi_Z(E) \in (0, 1]$ equal to the phase of the complex number $Z(E) \in \mathbb{H}$, so that

$$Z(E) = m(E)\, e^{i\pi\varphi(E)}, \qquad m(E) := |Z(E)| > 0.$$

The number $m(E)$ is called the *mass* of E.

Remark. In [MS17, Def. 4.1] one works with the generalized rank $R := \operatorname{Im} Z$ and the generalized degree $D := -\operatorname{Re} Z$; the phase corresponds to the generalized slope $M := D/R = -\cot(\pi\varphi) \in \mathbb{R} \cup \{+\infty\}$, with $M = +\infty$ for $\varphi = 1$. Since $\varphi \mapsto -\cot(\pi\varphi)$ is strictly increasing on $(0, 1)$, comparing slopes and comparing phases is the same thing: $M(F) \leqslant M(E)$ if and only if $\varphi(F) \leqslant \varphi(E)$, and likewise for strict inequalities. We shall mostly work with phases; the rank R, the degree D and the slope M reappear in the proof of existence of HN filtrations, where the geometry of the plane is more conveniently expressed through them.

Let Z be a stability function on $\mathcal{A}$. A nonzero object $E \in \mathcal{A}$ is called $(\mathcal{A}, Z)$-*semistable* ([MS17, Def. 4.2], [Bri07, Def. 2.2]) if

(AZ-SS) $\varphi(F) \leqslant \varphi(E)$ for every nontrivial subobject $F \subset E$,

and $(\mathcal{A}, Z)$-*stable* if

(AZ-S) $\varphi(F) < \varphi(E)$ for every nontrivial proper subobject $F \subset E$.

When $\mathcal{A}$ and Z are clear from the context we simply speak of semistable and stable objects.

1.3. **Harder–Narasimhan filtrations and stability conditions.** Let Z be a stability function on $\mathcal{A}$ and let $E \in \mathcal{A}$ be a nonzero object. A *Harder–Narasimhan filtration* (*HN filtration* for short) of E with respect to $(\mathcal{A}, Z)$ is a chain of subobjects

$$0 = E_0 \subset E_1 \subset \ldots \subset E_{n-1} \subset E_n = E$$

such that

- the factors $A_i := E_i/E_{i-1}$, $i = 1, \ldots, n$, are nonzero $(\mathcal{A}, Z)$-semistable objects;
- phases $\varphi(A_1) > \varphi(A_2) > \ldots > \varphi(A_n)$ are strictly decreasing.

The objects A_i are called the *HN factors* of E, and $\varphi^+(E) := \varphi(A_1)$, $\varphi^-(E) := \varphi(A_n)$ are called the *maximal* and the *minimal phase* of E.

$$\begin{array}{ccccccccccc} 0 = E_0 & \hookrightarrow & E_1 & \hookrightarrow & E_2 & \hookrightarrow & \cdots & \hookrightarrow & E_{n-1} & \hookrightarrow & E_n = E \\ & & \downarrow & & \downarrow & & & & \downarrow & & \downarrow \\ & & A_1 & & A_2 & & & & A_{n-1} & & A_n \end{array}$$

Each column $E_{i-1} \hookrightarrow E_i \twoheadrightarrow A_i$ of the diagram is a short exact sequence; the horizontal arrows are the inclusions of the filtration.

An $(\mathcal{A}, Z)$-semistable object E has the HN filtration $0 \subset E$ of length $n = 1$; conversely, if an HN filtration of E has length 1, then E is $(\mathcal{A}, Z)$-semistable. Uniqueness of HN filtrations will be proved below; until then the words "maximal phase" refer to a chosen filtration.

A *stability condition* on the abelian category $\mathcal{A}$ ([MS17, Def. 4.6], [Bri07, Def. 2.3]) is a pair $(\mathcal{A}, Z)$, where $Z\colon K_0(\mathcal{A}) \to \mathbb{C}$ is a group homomorphism, such that

(AZ-SC1) Z is a stability function on $\mathcal{A}$, i.e. condition (Z) holds;
(AZ-SC2) (*Harder–Narasimhan property*) every nonzero object $E \in \mathcal{A}$ admits an HN filtration with respect to $(\mathcal{A}, Z)$.

The homomorphism Z is called the *central charge* of the stability condition.

Remark. Condition (AZ-SC2) does not follow from (AZ-SC1): in Example 1.14 we shall give a stability function for which some object has no nonzero semistable subobject at all, hence no HN filtration. Sufficient conditions for the existence of HN filtrations (noetherianity of $\mathcal{A}$ together with discreteness of the image of $\operatorname{Im} Z$, or finite length of $\mathcal{A}$) will be discussed separately.

1.4. **Basic properties.** Throughout this subsection Z is a stability function on $\mathcal{A}$.

Lemma 1.1 (see-saw, [Rud97, Lemmas 1.2, 1.3])**.** *Let $0 = E_0 \subset E_1 \subset \ldots \subset E_m = E$ be a chain of subobjects with nonzero factors $B_j := E_j/E_{j-1}$, $j = 1, \ldots, m$. Then*

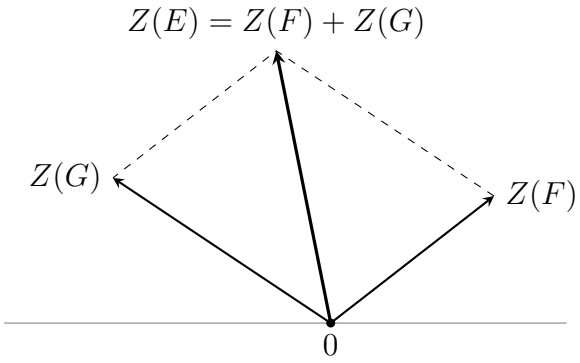
$$\min_j \varphi(B_j) \leqslant \varphi(E) \leqslant \max_j \varphi(B_j),$$

and equality holds on the left or on the right if and only if all the phases $\varphi(B_j)$ are equal.

In particular, for a short exact sequence $0 \to F \to E \to G \to 0$ with $F, G \neq 0$ the following are equivalent:

$$\varphi(F) < \varphi(E); \quad \varphi(E) < \varphi(G); \quad \varphi(F) < \varphi(G).$$

The same holds with $<$ replaced by $=$ or by $>$.



The picture shows the case of a short exact sequence $0 \to F \to E \to G \to 0$: since $Z(F)$ and $Z(G)$ both lie in $\mathbb{H}$, their sum $Z(E)$ lies in the sector between them, so the phase of E is caught between the phases of F and G.

Proof. By additivity, $Z(E) = \sum_j Z(B_j) = \sum_j r_j e^{i\pi\psi_j}$, where $r_j = m(B_j) > 0$ and $\psi_j = \varphi(B_j) \in (0,1]$. Put $\psi := \max_j \psi_j$. Then

$$e^{-i\pi\psi} Z(E) = \sum_j r_j e^{i\pi(\psi_j - \psi)}, \qquad \psi_j - \psi \in (-1, 0],$$

so every summand has non-positive imaginary part, with equality if and only if $\psi_j = \psi$, in which case the summand is the positive real number r_j. Hence $\operatorname{Im}\left(e^{-i\pi\psi} Z(E)\right) \leqslant 0$, with equality if and only if all ψ_j equal ψ. On the other hand $e^{-i\pi\psi} Z(E) = m(E) e^{i\pi\theta}$ with $\theta := \varphi(E) - \psi \in (-1, 1)$, and $\sin(\pi\theta) \leqslant 0$ on this interval exactly when $\theta \in (-1, 0]$, with $\sin(\pi\theta) = 0$ exactly when $\theta = 0$. This proves the upper bound and its equality case; the lower bound is obtained in the same way with $\min_j \psi_j$ in place of ψ.

For a short exact sequence apply the above to the chain $0 \subset F \subset E$ with factors F, G: if, say, $\varphi(F) \leqslant \varphi(G)$, then $\varphi(F) \leqslant \varphi(E) \leqslant \varphi(G)$ and $\varphi(E)$ equals one of the endpoints only if $\varphi(F) = \varphi(G)$; the case $\varphi(F) \geqslant \varphi(G)$ is symmetric. □

Lemma 1.2 (quotient characterization, [MS17, Exercise 4.4])**.** *A nonzero object E is $(\mathcal{A}, Z)$-semistable if and only if $\varphi(E) \leqslant \varphi(G)$ for every nonzero quotient $E \twoheadrightarrow G$, and $(\mathcal{A}, Z)$-stable if and only if $\varphi(E) < \varphi(G)$ for every nonzero proper quotient.*

Proof. Nontrivial proper subobjects $F \subset E$ correspond to nonzero proper quotients $G = E/F$, and by Lemma 1.1 the inequality $\varphi(F) \leqslant \varphi(E)$ is equivalent to $\varphi(E) \leqslant \varphi(G)$, and similarly for strict inequalities. The subobject $F = E$ and the quotient $G = E$ impose no condition. □

Lemma 1.3 (Schur, [Rud97, Thm 1], [MS17, Lemma 4.5])**.** *Let E, F be $(\mathcal{A}, Z)$-semistable objects.*

1. *If $\varphi(E) > \varphi(F)$, then* $\operatorname{Hom}(E, F) = 0$.
2. *If $\varphi(E) = \varphi(F)$ and $f\colon E \to F$ is nonzero, then f is a monomorphism provided E is stable, an epimorphism provided F is stable, and an isomorphism provided both are stable.*

Proof. Let $f\colon E \to F$ be nonzero, with image $I \neq 0$ and kernel K. Since I is a quotient of E, Lemma 1.2 gives $\varphi(E) \leqslant \varphi(I)$; since $I \subset F$, semistability of F gives $\varphi(I) \leqslant \varphi(F)$. Thus $\varphi(E) \leqslant \varphi(F)$, which proves (1). In the situation of (2) all these inequalities are equalities. If E is stable and $K \neq 0$, then K is a nontrivial proper subobject of E (proper because $I \neq 0$) and, by Lemma 1.1 applied to

$$0 \to K \to E \to I \to 0,$$

$\varphi(E) = \varphi(I)$ forces $\varphi(K) = \varphi(E)$, contradicting stability; so $K = 0$. If F is stable, then $I \subset F$ with $\varphi(I) = \varphi(F)$ forces $I = F$. The last claim combines the two. □

Lemma 1.4. *If E and F are $(\mathcal{A}, Z)$-semistable of the same phase φ, then so is $E \oplus F$.*

Proof. First,

$$Z(E \oplus F) = Z(E) + Z(F)$$

is a sum of two vectors on the ray $\mathbb{R}_{>0} e^{i\pi\varphi}$, so $\varphi(E \oplus F) = \varphi$. Let $G \subset E \oplus F$ be nontrivial, and put $G' := \ker(G \to F) = G \cap E \subset E$ and $G'' := \operatorname{Im}(G \to F) \subset F$, so that $0 \to G' \to G \to G'' \to 0$ is exact. If both G' and G'' are nonzero, they have

phases $\leqslant \varphi$ by semistability of E and F, and Lemma 1.1 gives $\varphi(G) \leqslant \varphi$; if one of them vanishes, G is isomorphic to the other and the same conclusion holds. □

Lemma 1.5. *Let E, F be nonzero objects admitting HN filtrations, with HN factors $A_1, \dots, A_n$ and $B_1, \dots, B_m$ respectively. If $\operatorname{Hom}(E, F) \neq 0$, then $\varphi^-(E) \leqslant \varphi^+(F)$; more precisely, $\varphi(A_i) \leqslant \varphi(B_j)$ for some i, j.*

Proof. Let $f\colon E \to F$ be nonzero, and let $0 = E_0 \subset \dots \subset E_n = E$, $0 = F_0 \subset \dots \subset F_m = F$ be the filtrations. Let i be minimal with $f|_{E_i} \neq 0$. Then $f|_{E_{i-1}} = 0$, so $f|_{E_i}$ factors through a nonzero map $g\colon A_i = E_i/E_{i-1} \to F$. Let j be minimal with $\operatorname{Im} g \subset F_j$. Then the composite $A_i \to F_j \to B_j = F_j/F_{j-1}$ is nonzero, since otherwise $\operatorname{Im} g \subset F_{j-1}$. By Lemma 1.3(1) applied to the semistable objects A_i and B_j, $\varphi(A_i) \leqslant \varphi(B_j)$. Finally $\varphi^-(E) \leqslant \varphi(A_i)$ and $\varphi(B_j) \leqslant \varphi^+(F)$. □

1.5. **Uniqueness of Harder–Narasimhan filtrations.**

Proposition 1.6 ([Rud97, Thm 2], [MS17, Exercise 4.7])**.** *Let Z be a stability function on $\mathcal{A}$ and let E be a nonzero object. If*

$$0 = E_0 \subset E_1 \subset \dots \subset E_n = E \qquad \text{and} \qquad 0 = E'_0 \subset E'_1 \subset \dots \subset E'_m = E$$

are two HN filtrations of E with respect to $(\mathcal{A}, Z)$, then $n = m$ and $E_i = E'_i$ as subobjects of E for all i. In particular the HN factors and the phases $\varphi^{\pm}(E)$ depend only on E.

Proof. Write $A_i := E_i/E_{i-1}$, $\varphi_i := \varphi(A_i)$ and $A'_j := E'_j/E'_{j-1}$, $\varphi'_j := \varphi(A'_j)$. We argue by induction on n.

Step 1 ($E_1 = E'_1$). Let j be minimal with $E_1 \subset E'_j$; it exists since $E'_m = E$. The composite $E_1 \to E'_j \to A'_j$ is nonzero, because otherwise $E_1 \subset E'_{j-1}$. Both $E_1 = A_1$ and A'_j are semistable, so Lemma 1.3(1) gives $\varphi_1 \leqslant \varphi'_j \leqslant \varphi'_1$. By symmetry $\varphi'_1 \leqslant \varphi_1$. Hence $\varphi_1 = \varphi'_1$, and then $\varphi'_j \geqslant \varphi'_1$ forces $j = 1$ because the phases φ'_j strictly decrease; thus $E_1 \subset E'_1$. By symmetry $E'_1 \subset E_1$, so $E_1 = E'_1$.

Step 2 (*induction*). If $n = 1$, then $E = E_1 = E'_1$, so $m = 1$ as well. If $n > 1$, consider $\bar{E} := E/E_1 = E/E'_1 \neq 0$. The images $\bar{E}_i := E_i/E_1$, $i \geqslant 1$, form a chain $0 = \bar{E}_1 \subset \dots \subset \bar{E}_n = \bar{E}$ with factors $\bar{E}_i/\bar{E}_{i-1} \cong E_i/E_{i-1} = A_i$ for $i \geqslant 2$, hence an HN filtration of $\bar{E}$ of length $n-1$; similarly for $\bar{E}'_j := E'_j/E'_1$. By induction $n - 1 = m - 1$ and $\bar{E}_i = \bar{E}'_i$ for all $i \geqslant 1$. Since subobjects of $\bar{E}$ correspond bijectively to subobjects of E containing $E_1 = E'_1$, we get $E_i = E'_i$ for all i. □

Corollary 1.7 ([Rud97, Prop. 1.9])**.** *Let $(\mathcal{A}, Z)$ be a stability condition and E a nonzero object. Then $\varphi^+(E) = \max\{\varphi(F) : 0 \neq F \subset E\}$, and the first step E_1 of the HN filtration is the unique largest subobject of E of phase $\varphi^+(E)$. Dually $\varphi^-(E) = \min\{\varphi(G) : E \twoheadrightarrow G \neq 0\}$.*

Proof. Let $0 \neq F \subset E$, with HN filtration $0 \subset F_1 \subset \dots \subset F$. The composite $F_1 \hookrightarrow F \hookrightarrow E$ is a nonzero map from the semistable object F_1 to E, so Lemma 1.5 gives $\varphi^+(F) = \varphi(F_1) \leqslant \varphi^+(E)$, and $\varphi(F) \leqslant \varphi^+(F)$ by Lemma 1.1. The value $\varphi^+(E)$ is attained by $F = E_1$. Now suppose $\varphi(F) = \varphi^+(E)$. Then $\varphi(F) = \varphi^+(F)$, so by the equality case of Lemma 1.1 all HN factors of F have phase $\varphi^+(E)$, i.e. F is semistable of phase $\varphi^+(E)$. All HN factors of E/E_1 have phase $< \varphi^+(E)$, so the composite

$F \to E \to E/E_1$ vanishes by Lemma 1.5, i.e. $F \subset E_1$. The dual statement follows by the same argument applied to quotients, using Lemma 1.2. □

1.6. **Existence of Harder–Narasimhan filtrations.** Recall that an abelian category $\mathcal{A}$ is *noetherian* if every ascending chain $F_1 \subset F_2 \subset \dots$ of subobjects of any object stabilizes. Equivalently, every nonempty family of subobjects of an object has a maximal element with respect to inclusion. Throughout this subsection Z is a stability function on $\mathcal{A}$, and we write $R := \operatorname{Im} Z$ and $D := -\operatorname{Re} Z$, so that $R(F) \geqslant 0$ for all F, and $R(F) = 0$ implies $D(F) > 0$ for $F \neq 0$. We say that R *has discrete image* if $R(K_0(\mathcal{A}))$ is a discrete subgroup of $\mathbb{R}$; in that case either $R \equiv 0$, or $R(K_0(\mathcal{A})) = c\mathbb{Z}$ for a unique $c > 0$, and then $R(E) \in c\mathbb{Z}_{\geqslant 0}$ for every object E.

Two elementary identities will be used repeatedly. If $F, F' \subset E$ are subobjects, the sequence

$$0 \to F \cap F' \to F \oplus F' \to F + F' \to 0$$

is exact, hence

$$Z(F + F') + Z(F \cap F') = Z(F) + Z(F'), \tag{1}$$

and the same holds for R and D separately. If $F \subset E$ and $R(F) = R(E)$, then $R(E/F) = 0$, so $D(E/F) \geqslant 0$ and

$$D(F) \leqslant D(E) \qquad \text{whenever } F \subset E \text{ and } R(F) = R(E). \tag{2}$$

Lemma 1.8 ([MS17, Lemma 4.9])**.** *Assume that*

- $\mathcal{A}$ *is noetherian and*
- *that* R *has discrete image in* $\mathbb{R}$.

Then for every object E *there is a constant* $D_E \in \mathbb{R}$ *such that* $D(F) \leqslant D_E$ *for every subobject* $F \subset E$.

Proof. If $R \equiv 0$, then every $F \subset E$ satisfies $R(F) = R(E)$ and (2) gives $D(F) \leqslant D(E)$. Otherwise write $R(K_0(\mathcal{A})) = c\mathbb{Z}$ with $c > 0$ and argue by induction on $k(E) := R(E)/c \in \mathbb{Z}_{\geqslant 0}$.

If $k(E) = 0$, then $R(F) = 0 = R(E)$ for every $F \subset E$, and (2) gives $D(F) \leqslant D(E)$.

Let $k(E) > 0$ and assume the lemma for all objects E' with $k(E') < k(E)$. Suppose, for a contradiction, that there are subobjects $F_n \subset E$, $n \in \mathbb{Z}_{>0}$, with $D(F_n) \to \infty$. By (2) only finitely many of them can satisfy $R(F_n) = R(E)$, so after discarding these we may assume $R(F_n) < R(E)$ for all n. We construct indices $n_1 < n_2 < \dots$ such that the subobjects $G_k := F_{n_1} + \dots + F_{n_k}$ satisfy

$$R(G_k) < R(E) \qquad \text{and} \qquad D(G_k) > D(G_{k-1}) \quad (k \geqslant 2).$$

Put $n_1 := 1$. Given $n_1 < \dots < n_{k-1}$, the object G_{k-1} satisfies $k(G_{k-1}) < k(E)$, so by the induction hypothesis there is a constant D' with $D(F') \leqslant D'$ for all $F' \subset G_{k-1}$. For $n > n_{k-1}$, identity (1) and $G_{k-1} \cap F_n \subset G_{k-1}$ give

$$D(G_{k-1} + F_n) = D(G_{k-1}) + D(F_n) - D(G_{k-1} \cap F_n) \geqslant D(G_{k-1}) + D(F_n) - D',$$

which tends to ∞ as $n \to \infty$. Choose $n_k > n_{k-1}$ with

$$D(G_{k-1} + F_{n_k}) > \max\{D(E), D(G_{k-1})\}.$$

Then $R(G_{k-1}+F_{n_k}) < R(E)$ by (2), and $D(G_k) > D(G_{k-1})$ as required. Since $D(G_k) > D(G_{k-1})$, the inclusion $G_{k-1} \subset G_k$ is strict, so $G_1 \subsetneq G_2 \subsetneq \dots$ is an infinite strictly ascending chain of subobjects of E, contradicting noetherianity. □

Proposition 1.9 ([MS17, Prop. 4.10]; cf. [Rud97, Thm 2], [Bri07, Prop. 2.4])**.** *Let Z be a stability function on $\mathcal{A}$. Assume that $\mathcal{A}$ is noetherian and that $R = \operatorname{Im} Z$ has discrete image. Then every nonzero object of $\mathcal{A}$ admits a Harder–Narasimhan filtration, i.e. $(\mathcal{A}, Z)$ is a stability condition.*

Proof. If $R \equiv 0$, then every nonzero object has phase 1, every nonzero object is semistable, and $0 \subset E$ is an HN filtration. So assume $R(K_0(\mathcal{A})) = c\mathbb{Z}$ with $c > 0$, and let $E \neq 0$.

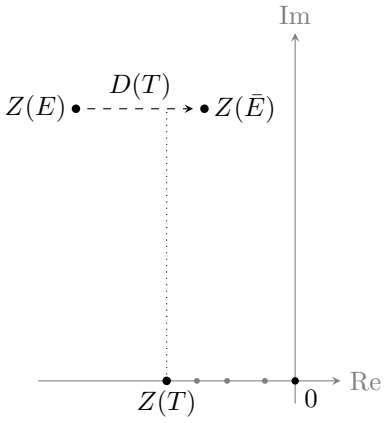
Step 1 (*the part of rank zero*). The family of subobjects $F \subset E$ with $R(F) = 0$ is nonempty (it contains 0), so it has a maximal element T. If $F \subset E$ is any subobject with $R(F) = 0$, then $R(T + F) = R(T) + R(F) - R(T \cap F) = -R(T \cap F) \leqslant 0$ by (1), hence $R(T + F) = 0$ and $T + F = T$ by maximality, i.e. $F \subset T$. Thus T is the largest subobject of E of rank zero. If $T \neq 0$, it is semistable of phase 1, because all its nonzero subobjects have $R = 0$, hence phase 1. Moreover the quotient $\bar{E} := E/T$ has no nonzero subobject of rank zero: the preimage $F \subset E$ of such a subobject $\bar{F}$ satisfies $R(F) = R(T) + R(\bar{F}) = 0$, hence $F \subset T$ and $\bar{F} = 0$. Consequently every nonzero subobject of $\bar{E}$ has $R \geqslant c$, hence phase < 1. If $\bar{E} = 0$, then $E = T$ is semistable and we are done. Otherwise, if $0 = \bar{E}_0 \subset \bar{E}_1 \subset \ldots \subset \bar{E}_m = \bar{E}$ is an HN filtration of $\bar{E}$, then its factors have phases < 1, and the preimages

$$0 \subset T \subset \pi^{-1}(\bar{E}_1) \subset \ldots \subset \pi^{-1}(\bar{E}_m) = E$$

form an HN filtration of E (with the first step omitted if $T = 0$), since the factors are T (of phase 1) followed by the factors of the filtration of $\bar{E}$. Hence we may and do assume from now on that

(3) every nonzero subobject $F \subset E$ satisfies $R(F) \geqslant c$.

In the pictures accompanying this proof we draw the plane of Z in the usual way, with $\operatorname{Re} Z$ to the right and $\operatorname{Im} Z$ upwards; a subobject $F \subset E$ is represented by the point $Z(F)$, whose height is the rank $R(F)$ and whose degree $D(F) = -\operatorname{Re} Z(F)$ grows to the left, so that the point with coordinates $(R, D) = (r, d)$ in the proof is the point $-d + ir$ of the plane. In Step 1, the subobjects of rank zero are the points on the negative real axis, T is the leftmost of them, and passing from E to $\bar{E} = E/T$ shifts every point to the right by $D(T)$.



Step 2 (*the Harder–Narasimhan polygon*). For $r \in c\mathbb{Z}$ with $0 \leqslant r \leqslant R(E)$ let $\mathcal{S}_r$ be the set of subobjects $F \subset E$ with $R(F) = r$, and call r *occupied* if $\mathcal{S}_r \neq \emptyset$. The set Λ_E of occupied values is finite and contains 0 and $R(E)$. For $r \in \Lambda_E$ put

$$d(r) := \sup\{D(F) : F \in \mathcal{S}_r\},$$

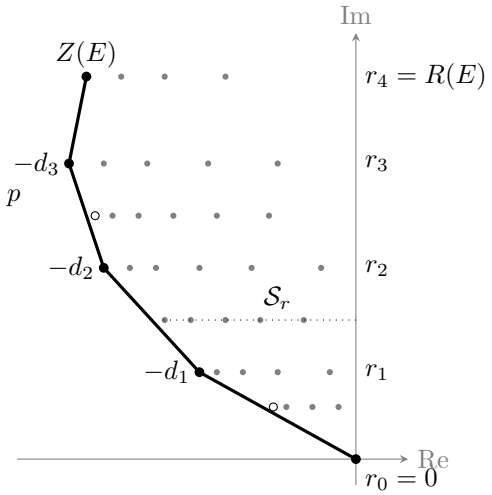
which is finite by Lemma 1.8. By (3) we have $\mathcal{S}_0 = \{0\}$ and $d(0) = 0$, and by (2) we have $d(R(E)) = D(E)$, attained by E itself. Let $p\colon [0, R(E)] \to \mathbb{R}$ be the upper concave envelope of the finite set of points $(r, d(r))$, $r \in \Lambda_E$, i.e. the smallest concave function with $p(r) \geqslant d(r)$ for all $r \in \Lambda_E$. It is piecewise linear, and its graph is the upper boundary of the convex hull of these points. Let $0 = r_0 < r_1 < \ldots < r_n = R(E)$ be the endpoints together with the points where the slope of p changes; we call them the *vertices*. They belong to Λ_E, satisfy $p(r_i) = d(r_i) =: d_i$, and the slopes

$$s_i := \frac{d_i - d_{i-1}}{r_i - r_{i-1}}, \qquad i = 1, \ldots, n,$$

are strictly decreasing, $s_1 > s_2 > \ldots > s_n$, because p is concave and changes slope at every r_i with $0 < i < n$. On $[r_{i-1}, r_i]$ the function p is affine with slope s_i. Finally, for every subobject $F \subset E$ we have $R(F) \in \Lambda_E$ and

$$D(F) \leqslant d(R(F)) \leqslant p(R(F)). \tag{4}$$

The picture below shows the sets $\mathcal{S}_r$ as horizontal rows of points at the occupied heights $r \in \Lambda_E$, the suprema $d(r)$ as the leftmost points of the rows (a hollow circle marks a supremum which is not attained), the graph of the envelope p as the thick line, and the vertices $r_0, \ldots, r_n$ as the heights at which this line bends; concavity of p means that the line bends to the right as one goes up.



Note that a non-vertex height may well be occupied (the row along the dotted line in the picture); what makes r_i a vertex is that the line bends there.

Step 3 (*the suprema at the vertices are attained*). We claim that for every vertex r_i there is $F_i \in \mathcal{S}_{r_i}$ with $D(F_i) = d_i$ (this is [MS17, Exercise 4.11]). For $i = 0$ take $F_0 = 0$ and for $i = n$ take $F_n = E$. Let $0 < i < n$ and choose a number s with $s_i > s > s_{i+1}$. Consider the linear functional $\ell(x, y) := y - sx$ on $\mathbb{R}^2$ and put $M := \ell(r_i, d_i) = d_i - sr_i$. Since s lies between the slopes of p to the left and to the right of r_i, concavity gives $p(x) \leqslant d_i + s(x - r_i)$ for all $x \in [0, R(E)]$, with equality only for $x = r_i$. Together with (4) this shows

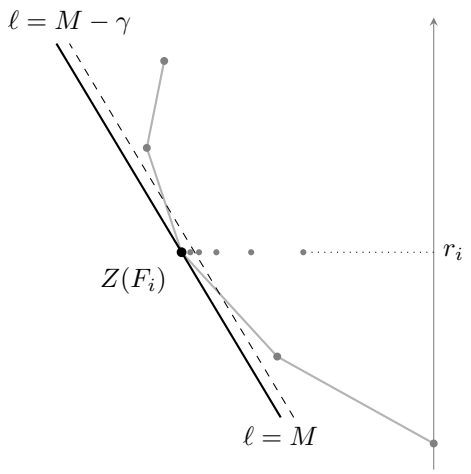
$$\ell\big(R(F), D(F)\big) \leqslant M \qquad \text{for every subobject } F \subset E, \tag{5}$$

and moreover, since Λ_E is finite,

$$\gamma := M - \max\big\{p(r) - sr : r \in \Lambda_E,\ r \neq r_i\big\} > 0,$$

so that $\ell(R(F), D(F)) > M - \gamma$ forces $R(F) = r_i$.

The picture shows the line through the vertex $Z(F_i) = -d_i + ir_i$ corresponding to the slope s: it is a supporting line of the polygon, touching it only at that vertex, so all points $Z(F)$ lie on it or to the right of it, and the points of the other rows stay to the right of the dashed parallel line, at horizontal distance γ from the first one; only the points of the row $\mathcal{S}_{r_i}$ can come arbitrarily close to the solid line.



For brevity write $\ell(F) := \ell(R(F), D(F))$. Applying ℓ to (1) and using (5) for $F \cap F'$ we get, for all $F, F' \subset E$,

$$\ell(F + F') = \ell(F) + \ell(F') - \ell(F \cap F') \geqslant \ell(F) + \ell(F') - M. \tag{6}$$

Now choose $F^{(1)}, F^{(2)}, \ldots \in \mathcal{S}_{r_i}$ with $D(F^{(j)}) \to d_i$, and put $\varepsilon_j := M - \ell(F^{(j)}) = d_i - D(F^{(j)}) \geqslant 0$. Passing to a subsequence we may assume $\sum_j \varepsilon_j < \gamma$. Let $G_k := F^{(1)} + \ldots + F^{(k)}$. By induction on k, (6) gives

$$\ell(G_k) \geqslant \sum_{j=1}^{k} \ell(F^{(j)}) - (k-1)M = M - \sum_{j=1}^{k} \varepsilon_j > M - \gamma.$$

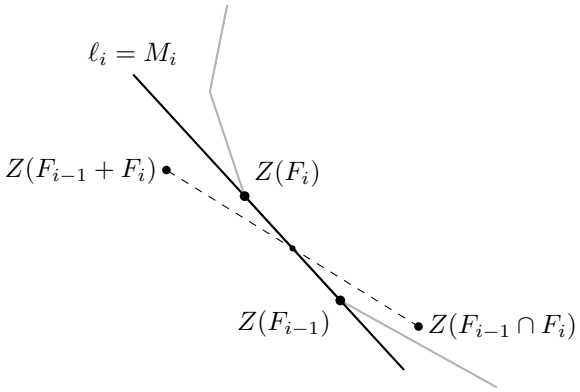
The chain $G_1 \subset G_2 \subset \ldots$ stabilizes by noetherianity, say $G_k = G$ for $k \geqslant N$. Then $\ell(G) > M - \gamma$, hence $R(G) = r_i$, and $G \supset F^{(j)}$ for every j. For $j \geqslant N$ the quotient $G/F^{(j)}$ has $R(G/F^{(j)}) = r_i - r_i = 0$, so $D(G/F^{(j)}) \geqslant 0$ and $D(G) \geqslant D(F^{(j)})$. Letting $j \to \infty$ we obtain $D(G) \geqslant d_i$, hence $D(G) = d_i$ by definition of d_i. Thus $F_i := G$ has the required property.

Step 4 (*the vertices form a chain*). Fix $1 \leqslant i \leqslant n$ and consider the functional $\ell_i(x, y) := y - s_i x$ and the number $M_i := \ell_i(r_{i-1}, d_{i-1}) = \ell_i(r_i, d_i)$. By concavity, $p(x) \leqslant d_{i-1} + s_i(x - r_{i-1})$ for all x, with equality exactly for $x \in [r_{i-1}, r_i]$, because the slope of p is $> s_i$ to the left of r_{i-1} and $< s_i$ to the right of r_i. Hence, by (4), $\ell_i(F) \leqslant M_i$ for every $F \subset E$, and $\ell_i(F) = M_i$ implies $r_{i-1} \leqslant R(F) \leqslant r_i$ and $D(F) = p(R(F))$. Apply (1) to F_{i-1} and F_i:

$$\ell_i(F_{i-1} \cap F_i) + \ell_i(F_{i-1} + F_i) = \ell_i(F_{i-1}) + \ell_i(F_i) = 2M_i.$$

Both summands on the left are $\leqslant M_i$, so both equal M_i.

In the picture, the points $Z(F_{i-1} \cap F_i)$ and $Z(F_{i-1} + F_i)$ are symmetric with respect to the midpoint of the edge from $Z(F_{i-1})$ to $Z(F_i)$ by (1), and both must lie on the line $\ell_i = M_i$ carrying that edge or to the right of it; the dashed configuration is therefore impossible, and both points are forced onto the line, hence onto the edge.



Since $R(F_{i-1} \cap F_i) \leqslant R(F_{i-1}) = r_{i-1}$, the first equality forces $R(F_{i-1} \cap F_i) = r_{i-1}$ and $D(F_{i-1} \cap F_i) = p(r_{i-1}) = d_{i-1} = D(F_{i-1})$. Therefore the quotient $F_{i-1}/(F_{i-1} \cap F_i)$ has $R = 0$ and $D = 0$, i.e. $Z = 0$, so it is zero by (Z). Hence $F_{i-1} \subset F_i$. We have obtained a chain

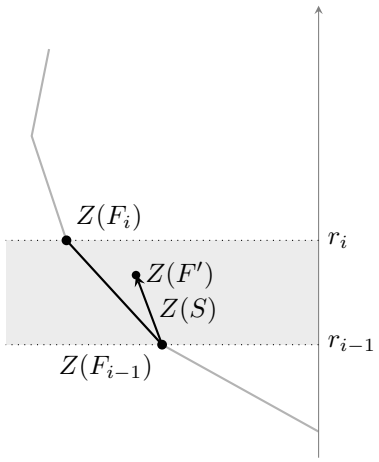
$$0 = F_0 \subset F_1 \subset \ldots \subset F_n = E.$$

Step 5 (*the factors are semistable with strictly decreasing phases*). Put $A_i := F_i/F_{i-1}$. Then $Z(A_i) = Z(F_i) - Z(F_{i-1})$, i.e. $R(A_i) = r_i - r_{i-1} > 0$ and $D(A_i) = d_i - d_{i-1}$, so $A_i \neq 0$ and its generalized slope is $M(A_i) = s_i$. Since $s_1 > \ldots > s_n$, the phases of the A_i strictly decrease. It remains to show that A_i is semistable. Let $S \subset A_i$ be a nontrivial subobject and $F' \subset F_i$ its preimage, so that $F_{i-1} \subsetneq F' \subset F_i$ and $Z(S) = Z(F') - Z(F_{i-1})$. Then $r_{i-1} \leqslant R(F') \leqslant r_i$, because $R(F'/F_{i-1}) \geqslant 0$ and $R(F_i/F') \geqslant 0$, and (4) together with the affine form of p on $[r_{i-1}, r_i]$ gives

$$D(F') \leqslant p(R(F')) = d_{i-1} + s_i\big(R(F') - r_{i-1}\big).$$

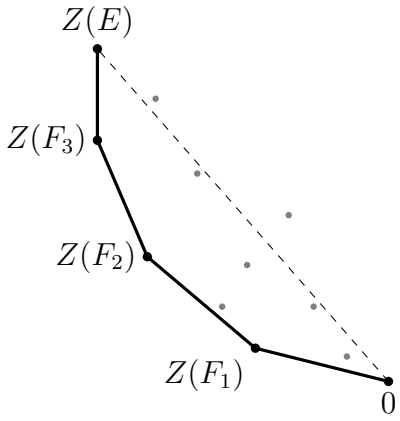
In other words $D(S) \leqslant s_i R(S)$.

In the picture, the point $Z(F')$ lies in the shaded horizontal strip $r_{i-1} \leqslant \mathrm{Im} \leqslant r_i$ and on the edge of the polygon or to the right of it; hence the vector from $Z(F_{i-1})$ to $Z(F')$, which is $Z(S)$ for $S = F'/F_{i-1}$, has slope at most s_i, i.e. phase at most $\varphi(A_i)$.



If $R(S) = 0$ this would give $D(S) \leqslant 0$, which is impossible for the nonzero object S; so $R(S) > 0$ and $M(S) = D(S)/R(S) \leqslant s_i = M(A_i)$, i.e. $\varphi(S) \leqslant \varphi(A_i)$. Thus A_i is semistable, and $0 = F_0 \subset \ldots \subset F_n = E$ is an HN filtration of E. □

The proof shows that the HN filtration can be read off from the *Harder–Narasimhan polygon* of E: the part of the boundary of the convex hull of $\{Z(F) : F \subset E\}$ lying to the left of the segment from 0 to $Z(E)$. Its vertices are the points $Z(F_i)$ and its edges are the vectors $Z(A_i)$. In the picture below the grey dots are the values $Z(F)$ of other subobjects $F \subset E$, the dashed segment joins 0 to $Z(E)$, and the thick polygonal line is the HN polygon.



Corollary 1.10 (Harder–Narasimhan [HN75])**.** *Let C be a smooth projective curve over a field $\Bbbk$ and $Z(E) = -\deg E + i \operatorname{rk} E$ on $\operatorname{Coh}(C)$. Then $(\operatorname{Coh}(C), Z)$ is a stability condition.*

Proof. $\operatorname{Coh}(C)$ is noetherian and $R = \operatorname{rk}$ takes values in $\mathbb{Z}$, so Proposition 1.9 applies. □

Remark. The proposition does not cover Example 1.13 in general, since there $\operatorname{Im} Z$ need not have discrete image; that example was treated by a direct finite-length argument. Bridgeland's original formulation [Bri07, Prop. 2.4] replaces the two hypotheses by two chain conditions on phases, which cover both situations at once; noetherianity excludes infinite descending chains of quotients with decreasing phases, and discreteness of R excludes infinite descending chains of subobjects with increasing phases.

1.7. **Examples.**

Example 1.11. Let C be a smooth projective curve over a field $\Bbbk$, $\mathcal{A} = \operatorname{Coh}(C)$, and

$$Z(E) := -\deg E + i \operatorname{rk} E.$$

This is a stability function: for a nonzero sheaf E either $\operatorname{rk} E > 0$, or E is a torsion sheaf, in which case $\deg E$ is the length of E, which is positive, so $Z(E) \in \mathbb{R}_{<0}$. For $\operatorname{rk} E > 0$ the generalized slope $M(E) = \deg E / \operatorname{rk} E$ is the usual slope, and torsion sheaves have $M = +\infty$; thus $(\mathcal{A}, Z)$-(semi)stability is slope (semi)stability, and semistable sheaves of positive rank are torsion-free. The Harder–Narasimhan property is Corollary 1.10,

originally due to Harder and Narasimhan [HN75]; the convex-polygon proof goes back to Shatz [Sha77]. So $(\mathrm{Coh}(C), Z)$ is a stability condition. ★

Example 1.12. Let X be a smooth projective variety of dimension $n \geqslant 2$ and ω an ample divisor class. The homomorphism $Z(E) := -\omega^{n-1}\cdot c_1(E) + i\,\omega^n \cdot \mathrm{rk}\,E$ on $K_0(\mathrm{Coh}(X))$ is not a stability function: for a nonzero sheaf T supported in codimension $\geqslant 2$ both $\mathrm{rk}\,T$ and $c_1(T)$ vanish, so $Z(T) = 0 \notin \mathbb{H}$. One can show that no homomorphism factoring through the Chern character is a stability function on $\mathrm{Coh}(X)$ (see [MS17, Example 4.3], where this is deduced from [Tod09, Lemma 2.7]); this is the reason for passing to other hearts in the derived category. ★

Example 1.13 ([MS17, Exercise 4.13])**.** Let A be a finite-dimensional algebra over a field $\Bbbk$, $\mathcal{A} = \mathrm{mod}\text{-}A$ the category of finite-dimensional right A-modules, and $S_1, \dots, S_m$ representatives of the isomorphism classes of simple modules. Every $M \in \mathcal{A}$ has a composition series, and by the Jordan–Hölder theorem the multiplicity $[M : S_j]$ of S_j among its factors is well defined and additive in short exact sequences. Hence

$$K_0(\mathcal{A}) \longrightarrow \mathbb{Z}^m, \qquad [M] \longmapsto \big([M : S_1], \dots, [M : S_m]\big)$$

is a well-defined homomorphism; it is surjective, and it is injective because every class $[M]$ equals $\sum_j [M : S_j][S_j]$. So $[S_1], \dots, [S_m]$ form a basis of $K_0(\mathcal{A}) \cong \mathbb{Z}^m$, and for any $z_1, \dots, z_m \in \mathbb{C}$ there is a unique homomorphism Z with $Z(S_j) = z_j$, namely $Z(M) = \sum_j [M : S_j] z_j$.

Now let $z_1, \dots, z_m \in \mathbb{H}$. Then Z is a stability function, since $Z(M)$ is a sum of vectors in $\mathbb{H}$ with non-negative coefficients, not all zero. We claim that $(\mathcal{A}, Z)$ is a stability condition. Let $M \neq 0$; we construct an HN filtration by induction on the length $\ell(M)$.

For a subobject $N \subset M$ one has $0 \leqslant [N : S_j] \leqslant [M : S_j]$, so the classes $[N]$, hence the phases $\varphi(N)$, range over a finite set. Let φ_1 be the maximal phase of a nontrivial subobject, and among the nontrivial subobjects of phase φ_1 choose M_1 of maximal length. Then M_1 is semistable: its subobjects are subobjects of M, hence have phase $\leqslant \varphi_1$. Moreover every nontrivial $N \subset M$ with $\varphi(N) = \varphi_1$ satisfies $N \subset M_1$. Indeed, from the exact sequence

$$0 \to M_1 \cap N \to M_1 \oplus N \to M_1 + N \to 0$$

and Lemma 1.4, $\varphi(M_1 \oplus N) = \varphi_1$, while $\varphi(M_1 \cap N) \leqslant \varphi_1$ if $M_1 \cap N \neq 0$; so Lemma 1.1 gives $\varphi(M_1 + N) \geqslant \varphi_1$, hence $= \varphi_1$ by maximality of φ_1. By maximality of $\ell(M_1)$, $\ell(M_1 + N) \leqslant \ell(M_1)$, so $M_1 + N = M_1$. (If $M_1 \cap N = 0$, then $M_1 + N \cong M_1 \oplus N$ has phase φ_1 and larger length, which is impossible, so this case does not occur.)

Finally, every nontrivial subobject $N' \subset M/M_1$ has $\varphi(N') < \varphi_1$: if $N \subset M$ is its preimage, then $0 \to M_1 \to N \to N' \to 0$ with $\varphi(N) \leqslant \varphi_1 = \varphi(M_1)$, so Lemma 1.1 gives $\varphi(N') \leqslant \varphi(N) \leqslant \varphi_1$, and $\varphi(N') = \varphi_1$ would force $\varphi(N) = \varphi_1$, hence $N \subset M_1$ and $N' = 0$. Since $\ell(M/M_1) < \ell(M)$, by induction M/M_1 has an HN filtration; its first factor has phase $< \varphi_1$ by what we just proved, so the preimage of this filtration in M, preceded by M_1, is an HN filtration of M. If $M_1 = M$ there is nothing to do. ★

Remark. The argument of Example 1.13 uses only that $\mathcal{A}$ has finite length and that the classes of subobjects of a given object form a finite set; it applies verbatim to any abelian category of finite length with finitely many simple objects.

Example 1.14 (a stability function without the Harder–Narasimhan property)**.** Let $\mathcal{A}$ be the category of finitely generated $\mathbb{Z}$-graded modules over the polynomial ring $\Bbbk[x]$ with $\deg x = 1$, and graded homomorphisms of degree 0. For $M = \bigoplus_n M_n \in \mathcal{A}$ the pieces M_n vanish for $n \ll 0$ and have bounded dimension, so

$$Z(M) := \sum_{n\in\mathbb{Z}} \big(-2^{-n} + i\,4^{-n}\big) \dim_{\Bbbk} M_n$$

converges absolutely. Short exact sequences in $\mathcal{A}$ are exact in each degree, so Z is additive, and $\operatorname{Im} Z(M) > 0$ for $M \neq 0$; thus Z is a stability function.

Consider $E := \Bbbk[x]$, generated in degree 0. Its graded submodules are exactly the ideals $(x^j) \cong \Bbbk[x](-j)$, $j \geqslant 0$, and 0. For $j \geqslant 0$,

$$Z\big((x^j)\big) = \sum_{n\geqslant j} \big(-2^{-n} + i\,4^{-n}\big) = -2^{1-j} + i\,\tfrac{4}{3}\,4^{-j},$$

so $\tan\big(\pi - \pi\varphi((x^j))\big) = \frac{2}{3}\cdot 2^{-j}$ is strictly decreasing in j; hence the phases $\varphi((x^j))$ strictly increase with j. Consequently no nonzero subobject of E is semistable: (x^j) has the subobject (x^{j+1}) of strictly larger phase. But the first step E_1 of an HN filtration of E would be a nonzero semistable subobject of E. Hence E has no HN filtration, and $(\mathcal{A}, Z)$ is not a stability condition. ★

2. t-structures and their hearts

2.1. **Conventions on triangulated categories.** Throughout this section $\mathcal{D}$ is an essentially small triangulated category with shift functor [1]. A *triangle* always means a distinguished triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$; we call Z a *cone* of f and write $Z = \operatorname{Cone}(f)$, keeping in mind that a cone is unique only up to (non-unique) isomorphism. We shall freely use the following standard facts, for which we refer to [GM03, Ch. IV].

(TR-rot) (*rotation*) If $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$ is a triangle, then so are $Y \xrightarrow{g} Z \xrightarrow{h} X[1] \xrightarrow{-f[1]} Y[1]$ and $Z[-1] \xrightarrow{-h[-1]} X \xrightarrow{f} Y \xrightarrow{g} Z$.

(TR-coh) (*Hom is cohomological*) For every object T and every triangle $X \to Y \to Z \to X[1]$ the sequences

$$\operatorname{Hom}(T, X) \to \operatorname{Hom}(T, Y) \to \operatorname{Hom}(T, Z),$$
$$\operatorname{Hom}(Z, T) \to \operatorname{Hom}(Y, T) \to \operatorname{Hom}(X, T)$$

are exact; combined with (TR-rot) this gives the long exact sequences $\ldots \to \operatorname{Hom}(T, Z[-1]) \to \operatorname{Hom}(T, X) \to \operatorname{Hom}(T, Y) \to \operatorname{Hom}(T, Z) \to \operatorname{Hom}(T, X[1]) \to \ldots$ and dually. In particular the composite of two consecutive maps in a triangle is zero.

(TR-5) (*five lemma*) If (α, β, γ) is a morphism of triangles and two of α, β, γ are isomorphisms, so is the third. In particular, in a triangle $0 \to Y \to Z \to 0$ the map $Y \to Z$ is an isomorphism.

(TR-oct) (*octahedral axiom*) For composable morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$ and chosen cones there is a triangle

$$\operatorname{Cone}(f) \to \operatorname{Cone}(gf) \to \operatorname{Cone}(g) \to \operatorname{Cone}(f)[1]$$

whose maps are compatible with the maps from Y, Z and to $X[1]$, $Y[1]$.

All subcategories are full and closed under isomorphisms. For a class of objects $\mathcal{S}$ we write

$$\mathcal{S}^{\perp} := \{Y \in \mathcal{D} |\, \mathrm{Hom}(S, Y) = 0 \text{ for all } S \in \mathcal{S}\},$$
$$^{\perp}\mathcal{S} := \{X \in \mathcal{D} |\, \mathrm{Hom}(X, S) = 0 \text{ for all } S \in \mathcal{S}\},$$

and $\mathcal{S}[n] := \{S[n] : S \in \mathcal{S}\}$. A subcategory $\mathcal{C}$ is *closed under extensions* if for every triangle $X \to Y \to Z \to X[1]$ with $X, Z \in \mathcal{C}$ also $Y \in \mathcal{C}$; the *extension closure* $\langle \mathcal{S} \rangle$ of a class $\mathcal{S}$ is the smallest subcategory containing $\mathcal{S}$ and 0 and closed under extensions. Since the split triangle $X \to X \oplus Z \to Z \xrightarrow{0} X[1]$ is distinguished, a subcategory closed under extensions is closed under finite direct sums.

2.2. **Definition.** A *t-structure* on $\mathcal{D}$ ([BBD82, Déf. 1.3.1]) is a pair $(\mathcal{D}^{\leqslant 0}, \mathcal{D}^{\geqslant 0})$ of subcategories of $\mathcal{D}$ such that, with the notation

$$\mathcal{D}^{\leqslant n} := \mathcal{D}^{\leqslant 0}[-n], \qquad \mathcal{D}^{\geqslant n} := \mathcal{D}^{\geqslant 0}[-n] \qquad (n \in \mathbb{Z}),$$

the following conditions hold:

(T1) $\mathcal{D}^{\leqslant -1} \subset \mathcal{D}^{\leqslant 0}$ and $\mathcal{D}^{\geqslant 1} \subset \mathcal{D}^{\geqslant 0}$, i.e. $\mathcal{D}^{\leqslant 0}[1] \subset \mathcal{D}^{\leqslant 0}$ and $\mathcal{D}^{\geqslant 0}[-1] \subset \mathcal{D}^{\geqslant 0}$;
(T2) $\mathrm{Hom}(X, Y) = 0$ for all $X \in \mathcal{D}^{\leqslant 0}$ and $Y \in \mathcal{D}^{\geqslant 1}$;
(T3) for every object $E \in \mathcal{D}$ there is a triangle

$$A \to E \to B \to A[1] \qquad \text{with } A \in \mathcal{D}^{\leqslant 0} \text{ and } B \in \mathcal{D}^{\geqslant 1}.$$

The subcategory $\mathcal{D}^{\leqslant 0}$ is called the *aisle* and $\mathcal{D}^{\geqslant 0}$ the *coaisle* of the t-structure. Applying the shift, (T1)–(T3) hold with 0 replaced by any $n \in \mathbb{Z}$: $\mathcal{D}^{\leqslant n-1} \subset \mathcal{D}^{\leqslant n}$, $\mathcal{D}^{\geqslant n+1} \subset \mathcal{D}^{\geqslant n}$, $\mathrm{Hom}(\mathcal{D}^{\leqslant n}, \mathcal{D}^{\geqslant n+1}) = 0$, and every object sits in a triangle $A \to E \to B \to A[1]$ with $A \in \mathcal{D}^{\leqslant n}$, $B \in \mathcal{D}^{\geqslant n+1}$. By induction, $\mathcal{D}^{\leqslant n} \subset \mathcal{D}^{\leqslant m}$ and $\mathcal{D}^{\geqslant m} \subset \mathcal{D}^{\geqslant n}$ for $n \leqslant m$.

The intuition comes from the standard example: for an abelian category $\mathcal{A}$ and $\mathcal{D} = \mathrm{D}^{\mathrm{b}}(\mathcal{A})$, the complexes with cohomology concentrated in degrees $\leqslant 0$, respectively $\geqslant 0$, form a t-structure, and the triangle in (T3) is the usual truncation

$$\tau^{\leqslant 0} E \to E \to \tau^{\geqslant 1} E$$

of complexes; this example will be treated in detail below.

2.3. **Truncation functors.**

Lemma 2.1 ([BBD82, Prop. 1.3.3])**.** *Let $(\mathcal{D}^{\leqslant 0}, \mathcal{D}^{\geqslant 0})$ satisfy (T1) and (T2), let $E \in \mathcal{D}$, and let $A \xrightarrow{a} E \xrightarrow{b} B \to A[1]$ be a triangle with $A \in \mathcal{D}^{\leqslant 0}$ and $B \in \mathcal{D}^{\geqslant 1}$.*

(1) *For every $T \in \mathcal{D}^{\leqslant 0}$ the map $a_* \colon \mathrm{Hom}(T, A) \to \mathrm{Hom}(T, E)$ is bijective, and for every $T \in \mathcal{D}^{\geqslant 1}$ the map $b^* \colon \mathrm{Hom}(B, T) \to \mathrm{Hom}(E, T)$ is bijective.*
(2) *If $A' \xrightarrow{a'} E \xrightarrow{b'} B' \to A'[1]$ is another triangle with $A' \in \mathcal{D}^{\leqslant 0}$ and $B' \in \mathcal{D}^{\geqslant 1}$, then there are unique morphisms $\alpha \colon A \to A'$ with $a'\alpha = a$ and $\beta \colon B \to B'$ with $\beta b = b'$; both are isomorphisms, and $(\alpha, \mathrm{Id}_E, \beta)$ is an isomorphism of triangles.*

Proof. (1) By (TR-coh) and (TR-rot) the sequence

$$\mathrm{Hom}(T, B[-1]) \to \mathrm{Hom}(T, A) \xrightarrow{a_*} \mathrm{Hom}(T, E) \to \mathrm{Hom}(T, B)$$

is exact. For $T \in \mathcal{D}^{\leqslant 0}$ the last term vanishes by (T2), and so does the first, because $B[-1] \in \mathcal{D}^{\geqslant 2} \subset \mathcal{D}^{\geqslant 1}$ by (T1). Hence a_* is bijective. Dually, the sequence

$$\mathrm{Hom}(A[1], T) \to \mathrm{Hom}(B, T) \xrightarrow{b^*} \mathrm{Hom}(E, T) \to \mathrm{Hom}(A, T)$$

is exact, and for $T \in \mathcal{D}^{\geqslant 1}$ both outer terms vanish, since $A[1] \in \mathcal{D}^{\leqslant -1} \subset \mathcal{D}^{\leqslant 0}$.

(2) Applying (1) to the second triangle with $T = A \in \mathcal{D}^{\leqslant 0}$ gives a unique $\alpha\colon A \to A'$ with $a'\alpha = a$; applying it to the first triangle with $T = A'$ gives a unique $\alpha'\colon A' \to A$ with $a\alpha' = a'$. Then $a(\alpha'\alpha) = a'\alpha = a = a\,\mathrm{Id}_A$, so $\alpha'\alpha = \mathrm{Id}_A$ by the uniqueness in (1), and likewise $\alpha\alpha' = \mathrm{Id}_{A'}$. In the same way there is a unique $\beta\colon B \to B'$ with $\beta b = b'$, and it is an isomorphism. By the axioms of a triangulated category the commutative square $a'\alpha = a\,\mathrm{Id}_E$ extends to a morphism of triangles $(\alpha, \mathrm{Id}_E, \beta'')$ for some $\beta''\colon B \to B'$; then $\beta'' b = b'$, so $\beta'' = \beta$ by uniqueness, and $(\alpha, \mathrm{Id}_E, \beta)$ is a morphism of triangles, which is an isomorphism by (TR-5). □

Let now $(\mathcal{D}^{\leqslant 0}, \mathcal{D}^{\geqslant 0})$ be a t-structure. For every object E we fix a triangle as in (T3) and write it

$$\tau^{\leqslant 0}E \to E \to \tau^{\geqslant 1}E \to (\tau^{\leqslant 0}E)[1]. \tag{7}$$

By Lemma 2.1(2) it is unique up to unique isomorphism. For a morphism $f\colon E \to E'$ there is a unique morphism $\tau^{\leqslant 0}f\colon \tau^{\leqslant 0}E \to \tau^{\leqslant 0}E'$ compatible with the maps to E and E', namely the preimage of $f \circ (\tau^{\leqslant 0}E \to E)$ under the bijection of Lemma 2.1(1) with $T = \tau^{\leqslant 0}E$; uniqueness implies $\tau^{\leqslant 0}(gf) = \tau^{\leqslant 0}g \circ \tau^{\leqslant 0}f$ and $\tau^{\leqslant 0}\mathrm{Id} = \mathrm{Id}$. Thus $\tau^{\leqslant 0}$ is a functor $\mathcal{D} \to \mathcal{D}^{\leqslant 0}$, and Lemma 2.1(1) says precisely that it is right adjoint to the inclusion $\mathcal{D}^{\leqslant 0} \hookrightarrow \mathcal{D}$. Dually $\tau^{\geqslant 1}\colon \mathcal{D} \to \mathcal{D}^{\geqslant 1}$ is a functor, left adjoint to the inclusion $\mathcal{D}^{\geqslant 1} \hookrightarrow \mathcal{D}$. For $n \in \mathbb{Z}$ put

$$\tau^{\leqslant n} := [-n] \circ \tau^{\leqslant 0} \circ [n], \qquad \tau^{\geqslant n} := [-n+1] \circ \tau^{\geqslant 1} \circ [n-1];$$

these are right, respectively left, adjoint to the inclusions of $\mathcal{D}^{\leqslant n}$ and $\mathcal{D}^{\geqslant n}$, and for every E there is a triangle $\tau^{\leqslant n}E \to E \to \tau^{\geqslant n+1}E \to (\tau^{\leqslant n}E)[1]$, unique up to unique isomorphism among triangles with first term in $\mathcal{D}^{\leqslant n}$ and third term in $\mathcal{D}^{\geqslant n+1}$. We call these the *truncation triangles* and $\tau^{\leqslant n}$, $\tau^{\geqslant n}$ the *truncation functors*.

Lemma 2.2 ([BBD82, 1.3.4])**.** *Let $(\mathcal{D}^{\leqslant 0}, \mathcal{D}^{\geqslant 0})$ be a t-structure. Then:*

1. *$\mathcal{D}^{\geqslant 1} = (\mathcal{D}^{\leqslant 0})^{\perp}$ and $\mathcal{D}^{\leqslant 0} = {}^{\perp}(\mathcal{D}^{\geqslant 1})$; in particular each of $\mathcal{D}^{\leqslant 0}$, $\mathcal{D}^{\geqslant 0}$ determines the other.*
2. *An object E lies in $\mathcal{D}^{\leqslant 0}$ if and only if $\tau^{\geqslant 1}E = 0$, and in $\mathcal{D}^{\geqslant 1}$ if and only if $\tau^{\leqslant 0}E = 0$.*
3. *$\mathcal{D}^{\leqslant 0}$ and $\mathcal{D}^{\geqslant 0}$ are closed under extensions, finite direct sums and direct summands.*

The same statements hold with 0 replaced by any $n \in \mathbb{Z}$.

Proof. (1) The inclusion $\mathcal{D}^{\geqslant 1} \subset (\mathcal{D}^{\leqslant 0})^{\perp}$ is (T2). Conversely let $E \in (\mathcal{D}^{\leqslant 0})^{\perp}$. The map $\tau^{\leqslant 0}E \to E$ is then zero, since $\tau^{\leqslant 0}E \in \mathcal{D}^{\leqslant 0}$. By Lemma 2.1(1) with $T = \tau^{\leqslant 0}E$ the map $\mathrm{Hom}(\tau^{\leqslant 0}E, \tau^{\leqslant 0}E) \to \mathrm{Hom}(\tau^{\leqslant 0}E, E)$ is injective, and it sends the identity to zero;

hence $\mathrm{Id}_{\tau^{\leqslant 0}E} = 0$, i.e. $\tau^{\leqslant 0}E = 0$. The truncation triangle becomes $0 \to E \to \tau^{\geqslant 1}E \to 0$, so $E \cong \tau^{\geqslant 1}E \in \mathcal{D}^{\geqslant 1}$ by (TR-5). The second equality is proved dually.

(2) If $E \in \mathcal{D}^{\leqslant 0}$, then $E \xrightarrow{\mathrm{Id}} E \to 0 \to E[1]$ is a triangle with first term in $\mathcal{D}^{\leqslant 0}$ and third term in $\mathcal{D}^{\geqslant 1}$, so $\tau^{\geqslant 1}E \cong 0$ by uniqueness. Conversely, if $\tau^{\geqslant 1}E = 0$, then $E \cong \tau^{\leqslant 0}E \in \mathcal{D}^{\leqslant 0}$. The second statement is dual.

(3) By (1), $\mathcal{D}^{\leqslant 0}$ is a left orthogonal ${}^{\perp}\mathcal{S}$ and $\mathcal{D}^{\geqslant 0} = \mathcal{D}^{\geqslant 1}[1]$ is a right orthogonal $\mathcal{S}'^{\perp}$. A left orthogonal is closed under extensions: if $X \to Y \to Z \to X[1]$ is a triangle with $\mathrm{Hom}(X, S) = \mathrm{Hom}(Z, S) = 0$, then $\mathrm{Hom}(Y, S) = 0$ by the exact sequence $\mathrm{Hom}(Z, S) \to \mathrm{Hom}(Y, S) \to \mathrm{Hom}(X, S)$ of (TR-coh); it is closed under direct summands because $\mathrm{Hom}(X \oplus X', S) = \mathrm{Hom}(X, S) \oplus \mathrm{Hom}(X', S)$; and closure under finite direct sums follows from closure under extensions. Right orthogonals are treated in the same way. □

Remark. Because of Lemma 2.2(1) a t-structure is often specified by its aisle $\mathcal{D}^{\leqslant 0}$ alone. Note also the shape of the truncation triangle in Lemma 2.1: it is exactly the triangle obtained by cutting a Harder–Narasimhan filtration between two consecutive factors, with $\mathcal{D}^{\leqslant 0}$ playing the role of "large phases" and $\mathcal{D}^{\geqslant 1}$ of "small phases"; this analogy will be made precise when we discuss slicings; see also the notion of t-stability in [GKR04].

2.4. **The heart.** Let $(\mathcal{D}^{\leqslant 0}, \mathcal{D}^{\geqslant 0})$ be a t-structure on $\mathcal{D}$. Its *heart* is the subcategory

$$\mathcal{A} := \mathcal{D}^{\leqslant 0} \cap \mathcal{D}^{\geqslant 0}.$$

By Lemma 2.2(3) it is closed under extensions, finite direct sums and direct summands, and it contains 0; so it is an additive subcategory of $\mathcal{D}$. For $n \in \mathbb{Z}$ we have $\mathcal{A}[n] = \mathcal{D}^{\leqslant -n} \cap \mathcal{D}^{\geqslant -n}$ (the objects "concentrated in degree $-n$"), and

$$\mathrm{Hom}(X, Y[k]) = 0 \qquad \text{for all } X, Y \in \mathcal{A} \text{ and } k < 0, \tag{8}$$

because $X \in \mathcal{D}^{\leqslant 0}$ and $Y[k] \in \mathcal{D}^{\geqslant -k} \subset \mathcal{D}^{\geqslant 1}$.

Throughout the rest of this subsection we fix the t-structure and its heart $\mathcal{A}$, and we repeatedly use the following observation.

Lemma 2.3. *Let $f\colon X \to Y$ be a morphism in $\mathcal{A}$ and $X \xrightarrow{f} Y \to C \to X[1]$ a triangle. Then $C \in \mathcal{D}^{\leqslant 0} \cap \mathcal{D}^{\geqslant -1}$, and in the truncation triangle $\tau^{\leqslant -1}C \to C \to \tau^{\geqslant 0}C \to (\tau^{\leqslant -1}C)[1]$ the objects*

$$K := (\tau^{\leqslant -1}C)[-1] \qquad \text{and} \qquad Q := \tau^{\geqslant 0}C$$

both lie in $\mathcal{A}$.

Proof. By (TR-rot), C is an extension of $X[1]$ by Y: there is a triangle

$$Y \to C \to X[1] \to Y[1].$$

Since $Y \in \mathcal{D}^{\leqslant 0}$ and $X[1] \in \mathcal{D}^{\leqslant -1} \subset \mathcal{D}^{\leqslant 0}$, Lemma 2.2(3) gives $C \in \mathcal{D}^{\leqslant 0}$; since $Y \in \mathcal{D}^{\geqslant 0}$ and $X[1] \in \mathcal{D}^{\geqslant -1}$, and $\mathcal{D}^{\geqslant 0} \subset \mathcal{D}^{\geqslant -1}$, it gives $C \in \mathcal{D}^{\geqslant -1}$.

By definition $\tau^{\leqslant -1}C \in \mathcal{D}^{\leqslant -1}$ and $Q = \tau^{\geqslant 0}C \in \mathcal{D}^{\geqslant 0}$. The rotated truncation triangle

$$Q[-1] \to \tau^{\leqslant -1}C \to C \to Q$$

exhibits $\tau^{\leqslant -1}C$ as an extension of $C \in \mathcal{D}^{\geqslant -1}$ by $Q[-1] \in \mathcal{D}^{\geqslant 1} \subset \mathcal{D}^{\geqslant -1}$, so $\tau^{\leqslant -1}C \in \mathcal{D}^{\geqslant -1}$ and hence $K = (\tau^{\leqslant -1}C)[-1] \in \mathcal{D}^{\leqslant 0} \cap \mathcal{D}^{\geqslant 0} = \mathcal{A}$. The triangle

$$C \to Q \to (\tau^{\leqslant -1}C)[1] \to C[1]$$

exhibits Q as an extension of $(\tau^{\leqslant -1}C)[1] \in \mathcal{D}^{\leqslant -2} \subset \mathcal{D}^{\leqslant 0}$ by $C \in \mathcal{D}^{\leqslant 0}$, so $Q \in \mathcal{D}^{\leqslant 0}$ and hence $Q \in \mathcal{A}$. □

In the situation of Lemma 2.3 we define morphisms $\kappa\colon K \to X$ and $\pi\colon Y \to Q$ as the composites

$$\kappa\colon K = (\tau^{\leqslant -1}C)[-1] \longrightarrow C[-1] \longrightarrow X, \qquad \pi\colon Y \longrightarrow C \longrightarrow Q,$$

where $C[-1] \to X$ is the map of the rotated triangle $C[-1] \to X \xrightarrow{f} Y \to C$.

Proposition 2.4. *In the situation of Lemma 2.3, $\kappa\colon K \to X$ is a kernel of f in $\mathcal{A}$ and $\pi\colon Y \to Q$ is a cokernel of f in $\mathcal{A}$.*

Proof. We show that for every $T \in \mathcal{A}$ the sequence

$$0 \to \operatorname{Hom}(T, K) \xrightarrow{\kappa_*} \operatorname{Hom}(T, X) \xrightarrow{f_*} \operatorname{Hom}(T, Y)$$

is exact; by the Yoneda lemma this means that κ is a kernel of f in $\mathcal{A}$.

First, $\operatorname{Hom}(T, K) \to \operatorname{Hom}(T, C[-1])$ is bijective. Indeed, the triangle

$$K \to C[-1] \to Q[-1] \to K[1]$$

(the truncation triangle of C shifted by $[-1]$) gives an exact sequence

$$\operatorname{Hom}(T, Q[-2]) \to \operatorname{Hom}(T, K) \to \operatorname{Hom}(T, C[-1]) \to \operatorname{Hom}(T, Q[-1]),$$

and both outer terms vanish by (T2), because $T \in \mathcal{D}^{\leqslant 0}$ while $Q[-1] \in \mathcal{D}^{\geqslant 1}$ and $Q[-2] \in \mathcal{D}^{\geqslant 2} \subset \mathcal{D}^{\geqslant 1}$.

Next, the rotated triangle $Y[-1] \to C[-1] \to X \xrightarrow{f} Y$ gives an exact sequence

$$\operatorname{Hom}(T, Y[-1]) \to \operatorname{Hom}(T, C[-1]) \to \operatorname{Hom}(T, X) \xrightarrow{f_*} \operatorname{Hom}(T, Y),$$

in which $\operatorname{Hom}(T, Y[-1]) = 0$ by (8). Hence $\operatorname{Hom}(T, C[-1]) \to \operatorname{Hom}(T, X)$ is injective with image $\ker f_*$. Composing with the bijection of the previous paragraph gives the claim for $\kappa_* = (\text{this map}) \circ (\text{bijection})$.

The statement about π is proved dually, using the exact sequences

$$\operatorname{Hom}(K[2], T) \to \operatorname{Hom}(Q, T) \to \operatorname{Hom}(C, T) \to \operatorname{Hom}(K[1], T),$$

$$\operatorname{Hom}(X[1], T) \to \operatorname{Hom}(C, T) \to \operatorname{Hom}(Y, T) \xrightarrow{f^*} \operatorname{Hom}(X, T),$$

in which $\operatorname{Hom}(K[1], T) = \operatorname{Hom}(K[2], T) = 0$ by (T2) (as $K[1] \in \mathcal{D}^{\leqslant -1}$, $K[2] \in \mathcal{D}^{\leqslant -2}$ and $T \in \mathcal{D}^{\geqslant 0}$, so $\operatorname{Hom}(K[j], T) = \operatorname{Hom}(K[j-1], T[-1])$ with $T[-1] \in \mathcal{D}^{\geqslant 1}$) and $\operatorname{Hom}(X[1], T) = 0$ by (8). □

Lemma 2.5. *Let $X', X, X'' \in \mathcal{A}$.*

(1) *If $X' \xrightarrow{u} X \xrightarrow{v} X'' \to X'[1]$ is a triangle, then $0 \to X' \xrightarrow{u} X \xrightarrow{v} X'' \to 0$ is a short exact sequence in $\mathcal{A}$, i.e. u is a kernel of v and v is a cokernel of u.*

(2) *Conversely, if* $0 \to X' \xrightarrow{u} X \xrightarrow{v} X'' \to 0$ *is a short exact sequence in* $\mathcal{A}$ *(meaning that* u *is a kernel of* v *and* v *is a cokernel of* u*), then there is a unique morphism* $w\colon X'' \to X'[1]$ *such that* $X' \xrightarrow{u} X \xrightarrow{v} X'' \xrightarrow{w} X'[1]$ *is a triangle.*

Proof. (1) Apply Lemma 2.3 and Proposition 2.4 to $f = u$ with cone $C = X''$. Since $X'' \in \mathcal{D}^{\geqslant 0}$, Lemma 2.2(2) (shifted) gives $\tau^{\leqslant -1}X'' = 0$, so $K = 0$ and $Q = X''$ with $\pi = v$; thus u is a monomorphism (its kernel is zero) and v is a cokernel of u. Apply them next to $f = v$, whose cone is $X'[1]$ by (TR-rot): since $X'[1] \in \mathcal{D}^{\leqslant -1}$ we get $\tau^{\leqslant -1}(X'[1]) = X'[1]$ and $\tau^{\geqslant 0}(X'[1]) = 0$, so $K = X'$ and $Q = 0$; thus v is an epimorphism and its kernel is $\kappa\colon X' \to X$, which is u up to sign (the sign comes from the rotation in (TR-rot) and does not affect the kernel).

(2) Complete u to a triangle

$$X' \xrightarrow{u} X \xrightarrow{v'} C \to X'[1]$$

and form K, Q, κ, π as in Lemma 2.3. By Proposition 2.4, $\kappa\colon K \to X'$ is a kernel of u; as u is a monomorphism, $K = 0$, i.e. $\tau^{\leqslant -1}C = 0$, and the map $C \to Q = \tau^{\geqslant 0}C$ is an isomorphism by (TR-5). In particular $C \in \mathcal{A}$, and $\pi\colon X \to Q$ identifies with $v'\colon X \to C$, which is therefore a cokernel of u. Two cokernels of u are related by a unique isomorphism $\theta\colon C \to X''$ with $\theta v' = v$. Transporting the triangle along θ we obtain a triangle

$$X' \xrightarrow{u} X \xrightarrow{v} X'' \xrightarrow{w} X'[1].$$

For uniqueness of w: if w' also completes u, v to a triangle, then by the axioms there is a morphism of triangles $(\mathrm{Id}, \mathrm{Id}, \gamma)$ between the two, with $\gamma \in \mathrm{End}(X'')$ satisfying $\gamma v = v$, hence $\gamma = \mathrm{Id}$ since v is an epimorphism; the compatibility with the third maps then reads $w' = w$. □

Theorem 2.6 ([BBD82, Théorème 1.3.6])**.** *The heart* $\mathcal{A}$ *of a t-structure is an abelian category. Its short exact sequences are exactly the triangles of* $\mathcal{D}$ *with all three terms in* $\mathcal{A}$*, in the sense of Lemma 2.5.*

Proof. $\mathcal{A}$ is additive, and every morphism has a kernel and a cokernel by Proposition 2.4. It remains to show that for every $f\colon X \to Y$ in $\mathcal{A}$ the canonical morphism from the coimage $\mathrm{coker}(\ker f \to X)$ to the image $\ker(Y \to \mathrm{coker}\, f)$ is an isomorphism.

Let $X \xrightarrow{f} Y \xrightarrow{g} C \to X[1]$ be a triangle, and let K, Q, κ, π be as above, with $q\colon C \to Q$ the truncation map, so that $\pi = qg$. Choose a triangle

$$I \xrightarrow{j} Y \xrightarrow{\pi} Q \to I[1].$$

We apply (TR-oct) to the composable pair $Y \xrightarrow{g} C \xrightarrow{q} Q$: the cones are $\mathrm{Cone}(g) = X[1]$, $\mathrm{Cone}(qg) = \mathrm{Cone}(\pi) = I[1]$ and $\mathrm{Cone}(q) = (\tau^{\leqslant -1}C)[1] = K[2]$, so there is a triangle

$$X[1] \to I[1] \to K[2] \to X[2],$$

which after rotating and shifting becomes a triangle

$$K \to X \xrightarrow{e} I \to K[1];$$

the compatibilities in the octahedral axiom say that $je = f$ and that $K \to X$ is the map κ (up to sign).

We claim that $I \in \mathcal{A}$. From the triangle

$$Q[-1] \to I \xrightarrow{j} Y \to Q$$

and $Q[-1] \in \mathcal{D}^{\geqslant 1} \subset \mathcal{D}^{\geqslant 0}$, $Y \in \mathcal{D}^{\geqslant 0}$ we get $I \in \mathcal{D}^{\geqslant 0}$; from the triangle $X \xrightarrow{e} I \to K[1] \to X[1]$ and $X \in \mathcal{D}^{\leqslant 0}$, $K[1] \in \mathcal{D}^{\leqslant -1} \subset \mathcal{D}^{\leqslant 0}$ we get $I \in \mathcal{D}^{\leqslant 0}$.

Now Lemma 2.5(1) applies to both triangles

$$K \to X \xrightarrow{e} I \to K[1]$$

and $I \xrightarrow{j} Y \xrightarrow{\pi} Q \to I[1]$, all of whose terms lie in $\mathcal{A}$:

$$0 \to K \xrightarrow{\pm\kappa} X \xrightarrow{e} I \to 0 \qquad \text{and} \qquad 0 \to I \xrightarrow{j} Y \xrightarrow{\pi} Q \to 0$$

are short exact sequences in $\mathcal{A}$. The first says that e is a cokernel of $\kappa = \ker f$, i.e. $e\colon X \to I$ is the coimage of f; the second says that j is a kernel of $\pi = \operatorname{coker} f$, i.e. $j\colon I \to Y$ is the image of f. Since $f = je$, the canonical map $\operatorname{coim} f \to \operatorname{im} f$ is the identity of I, an isomorphism. Hence $\mathcal{A}$ is abelian. The last assertion is Lemma 2.5. □

Remark. The proof shows more precisely how kernels, cokernels and images of a morphism $f\colon X \to Y$ in $\mathcal{A}$ are read off from its cone C: the cone lives in two consecutive "degrees" -1 and 0, and

$$\ker f = (\tau^{\leqslant -1}C)[-1], \qquad \operatorname{coker} f = \tau^{\geqslant 0}C.$$

In particular f is a monomorphism in $\mathcal{A}$ if and only if $C \in \mathcal{A}$, and an epimorphism if and only if $C \in \mathcal{A}[1]$. Note also that a monomorphism in $\mathcal{A}$ need not be a monomorphism in $\mathcal{D}$, and the heart is in general not closed under cones: if f has nonzero kernel, then $C \notin \mathcal{A}$.

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